Zachary D. Calhoun
Duke University
Pratt School of Engineering
Department of Civil & Environmental Engineering
534 Research Drive, Wilkinson Building, Room 436
Durham, North Carolina 27705
zachary.calhoun@duke.edu | c. (404) 312-9998

EDUCATION

Exp. 2026 **Duke University**
Ph.D., Civil & Environmental Engineering

2023 **Duke University**
M.S., Civil & Environmental Engineering

2017 **University of Virginia**
B.S., Systems & Information Engineering, 2nd Major Spanish Language

RESEARCH EXPERIENCE

2021-present **Graduate Research Assistant**
Duke University, Department of Civil & Environmental Engineering
Advisors: David Carlson & Michael Bergin

- Developed new research direction studying applications of machine learning, spatial statistics, and causal inference to model urban heat stress and interventions.
- Led three peer-reviewed publications, with three more currently under review.
- Presented work at AGU, JSM, and ICUC.

PUBLICATIONS

Refereed

2025 Li, K., Wood, C., Nichols, L., **Calhoun, Z. D.**, Bhavsar, N. A., Carlson, D. Neighborhood Temperature and Contextual Factors Improve Prediction of Childhood Body Mass Index: An Application of Novel Graph Neural Networks, *AJE Advances: Research in Epidemiology,* 1, 8. https://doi.org/10.1093/ajeadv/uuaf011.

2024 **Calhoun, Z. D.**, Black, M. S., Bergin, M. & Carlson, D. Refining citizen climate science: Addressing preferential sampling for improved estimates of urban heat. *Environmental Science & Technology Letters*, 11, 8. https://doi.org/10.1021/acs.estlett.4c00296.

2024 **Calhoun, Z. D.**, Willard, F., Ge, C., Rodriguez, C., Bergin, M. & Carlson, D. Estimating the effects of vegetation and increased albedo on the urban heat island effect with spatial causal inference. *Scientific Reports,* 14, 540. https://doi.org/10.1038/s41598-023-50981-w.

2022 **Calhoun, Z. D.**, Lahrichi, S., Ren, S., Malof, J. M. & Bradbury, K. Self-supervised encoders are better transfer learners in remote sensing applications. *Remote Sensing,* 14, 21. https://doi.org/10.3390/rs14215500.

Under review (date denotes submission, * denotes equal contribution)

5/2026 Kim, E. K., **Calhoun, Z. D.,** Bergin, M., & Carlson D. Mapping urban heat from citizen-collected data: a scalable Gaussian process approach to correcting preferential sampling. (under review at *Environmental Data Science).*

1/2026 Bajgain, T. R.*, **Calhoun, Z. D.***, Tripathi, S. N., Salam, A., Jain, V., Zaman, S. U., Roy, S., Bergin, M., & Carlson, D. How many $PM_{2.5}$ sensors are needed to accurately monitor city-wide daily and annual mean concentrations? (under review at *ES&T Air*)

9/2025 **Calhoun, Z. D.**, Bergin, M., & Carlson, D. Probabilistic interpolation of crowdsourced meteorological data for higher-resolution gridded estimates of surface air temperature. (under review at *Urban Climate,* pre-print available at https://doi.org/10.31223/X5B45S).

Conference Papers

2022 **Calhoun, Z. D.**, Jiang, Z., Bergin, M. & Carlson, D. Urban heat island detection and causal inference using convolutional neural networks, *Proceedings of the 2022 NeurIPS Workshop on Climate Change.*

2017 **Calhoun, Z. D.,** Maribojoc, P., Selzer, N., Procopi, L., Bezzo, N. & Fleming, C. Analysis of identity and access management alternatives for a multinational information-sharing environment, *Proceedings of the 2017 Systems and Information Engineering Design Symposium.*

2016 Huzaifa, U., Bernier, C., **Calhoun, Z. D.**, Heddy, G., Kohout, C., Libowitz, B., Moenning, A., Ye, J., Maguire, C. & LaViers, A. Embodied movement strategies for development of a core-located actuation walker, *Proceedings of the 6th IEEE International Conference on Biomedical Robotics and Biomechatronics.*

PRESENTATIONS

Invited Talks

10/2025 **Calhoun, Z. D.** Measuring and Mitigating Urban Heat: Statistical Tools for Climate Adaptation. Presented to the Earth and Atmospheric Sciences Department at Georgia Tech. Atlanta, GA.

10/2024 **Calhoun, Z. D.** Matching Heat Maps to Health Outcomes: Urban Heat Stress and Opportunities to Intervene. Underwriters Laboratories Research Symposium. Atlanta, GA.

8/2024 Black, M., Azan, A., Chepaitis, P., **Calhoun, Z. D.** Building Resilience for Health: Exploring the Impact of Extreme Heat on Human Health. Webinar hosted by Underwriters Laboratories: Chemical Insights Research Institute. Online.

8/2023 **Calhoun, Z. D.** & Bergin, M. The Changing Climate and Impact on Air Pollution. Underwriters Laboratories Research Symposium. Evanston, IL.

Conference Presentations (Oral)

8/2025 **Calhoun, Z. D.,** Kim, E., Bergin, M., Carlson, D. Modeling urban heat stress with preferentially sampled citizen science data, *2025 Joint Statistical Meetings*, Nashville, TN.

4/2025 **Calhoun, Z. D.** Big, noisy data: how scalable Gaussian processes can leverage personal weather stations to improve spatiotemporal coverage of urban climate networks, *Duke SympoCEEum*, Durham, NC. **(Best Oral Presentation Award)**

12/2024 **Calhoun, Z. D.,** Black, M.S., Bergin, M., Carlson, D. Spatiotemporal estimates of urban heat stress using citizen science: how adjusting for preferentially sampled observations can fill in the gaps, *American Geophysical Union (AGU) Fall Meeting,* Washington, DC.

Conference Presentations (Poster)

7/2025 **Calhoun, Z. D.**, Bergin, M., Carlson, D. Assimilating crowdsourced weather data into urban climate models: a fast geostatistical framework using scalable Gaussian processes. *12th International Conference on Urban Climate.* Rotterdam, Netherlands.

5/2025 **Calhoun, Z. D.**, Jiang, G., Willard, F., Ge, C., Rodriguez, C., Bergin, M., & Carlson, D. Towards precision urban climate: estimating the effects of interventions on the urban heat island effect with spatial causal inference. *Association of Environmental Engineering and Science Professors 2025 Conference,* Durham, NC. **(Best Poster Award)**

10/2024 Liao, E., **Calhoun, Z. D.**, Bergin, M., & Carlson, D. Urban heat mitigation: Predicting the efficacy of heat-reducing interventions using spatial causal inference. *9th Annual Clean AIRE NC Breathe Conference,* Charlotte, NC. **(Best Poster Award)**

5/2022	**Calhoun, Z. D.**, Bergin, M., & Carlson, D. Measuring the health effects of severe air pollution incidents using spatiotemporally tagged tweets. *Workshop on AI and Open Data Practices in Chemical Hazard Assessment. National Academies of Sciences, Engineering, and Medicine.* Online.

AWARDS & FELLOWSHIPS

2026	Bass Teaching Assistantship Fellowship, Duke University • Awarded $16,000 to support introductory undergraduate computing course.
2025, 2023, 2022	Climate+ Summer Research Projects, Duke University • Won funding to support team of undergraduate students for three separate projects.
2024	Utku Best Pre-PhD Paper Award
2021	Duke Pratt-Gardner Graduate Fellowship

TEACHING & MENTORSHIP

Teaching Assistantships

S' 2024	Air Pollution Engineering, Duke University
F' 2023	Data Science and Machine Learning for Engineers, Duke University

Guest Lectures

S' 2026	Building an academic website with AI tools. Presented to the Department of Biostatistics and Bioinformatics at Duke University.
S' 2025	Air Pollution Engineering, Duke University

Undergraduate Independent Study Mentorship

2025-present	Ella Tallett, Duke University, incoming Ph.D. student at WashU
2025-present	Oumaima Berrada, Duke Kunshan University, incoming Masters student at Oxford
2024-2025	Edrian Liao, Duke University, currently Ph.D. student at MIT
2024	Daniel Oren, Duke University
2023	Frank Willard, Duke University

Undergraduate Summer Research Mentorship

2025	"Adapting to warmer temperatures at the city-scale" Students: Oumaima Berrada, Ella Tallett, Rittik Barua, Yazlin Moujalled
2023	"Measuring urban heat islands and their causes in Durham" Students: Chenhao Ge, Claudia Rodriguez
2022	"Tracking climate change causes & impacts with satellites and AI" Students: Saad Lahrichi, Rebecca Lan, Edrian Liao

TRAINING & CERTIFICATES

2024-present	College Teaching Certificate, Duke University

INDUSTRY EXPERIENCE

2019-2021	**Peloton Interactive, New York, NY** Supply Chain IT Analyst • Designed systems to track inventory from manufacturing to home delivery.
2017-2019	**Myers-Holum, Inc., New York, NY** Technical Consultant & Developer • Developed project budgeting and accounting software for clients.

TECHNICAL SKILLS

Programming languages: Python, R, Bash, JavaScript, SQL
Libraries: PyTorch, GPyTorch, geopandas, rasterio, xarray, GDAL
Software: Google Earth Engine, Slurm

SERVICE

University Service

2023-present **PhD+ Executive Board Member, Duke University**
Organized events with invited speakers to discuss career options for PhD students in academia, government, and industry.

Journals refereed for

- *ARC Geophysical Research*
- *Geohealth*
- *Earth Science Informatics*
- *Nature Scientific Reports*
- *Remote Sensing Applications: Society and Environment*

PROFESSIONAL AFFILIATIONS

2025-present	Association of Environmental Engineering and Science Professors
2025-present	American Statistical Assocation
2024-present	International Association for Urban Climate
2024	American Geophysical Union